

Program 7 : Deploy web application to k8s using minikube

```
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg7$ cat nginx-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest # already exists on Docker Hub
          ports:
            - containerPort: 80
```

```
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg7$ minikube start
🐳 minikube v1.37.0 on Ubuntu 22.04
🔧 Automatically selected the docker driver
👉 Using Docker driver with root privileges
🌟 Starting "minikube" primary control-plane node in "minikube" cluster
📦 Pulling base image v0.0.48 ...
📦 Downloading Kubernetes v1.34.0 preload ...
> gcr.io/k8s-minikube/kicbase...: 488.51 MiB / 488.52 MiB 100.00% 6.18 Mi
> preloaded-images-k8s-v18-v1...: 337.07 MiB / 337.07 MiB 100.00% 3.32 Mi
🔥 Creating docker container (CPUs=2, Memory=3900MB) ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔧 Configuring bridge CNI (Container Networking Interface) ...
🔧 Verifying Kubernetes components...
   ■ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟 Enabled addons: storage-provisioner, default-storageclass
🎉 Done! kubectrl is now configured to use "minikube" cluster and "default" namespace by default
```

```
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg7$ minikube status
```

```
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

```
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg7$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-h8wjc   0/1     ContainerCreating   0          11s
nginx-deployment-6f9664446b-xpszz   0/1     ContainerCreating   0          11s
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg7$ kubectl get nodes
NAME      STATUS   ROLES    AGE   VERSION
minikube  Ready    control-plane   98s   v1.34.0
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg7$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-h8wjc   0/1     ContainerCreating   0          19s
nginx-deployment-6f9664446b-xpszz   1/1     Running            0          19s
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg7$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-h8wjc   1/1     Running    0          34s
nginx-deployment-6f9664446b-xpszz   1/1     Running    0          34s
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg7$ kubectl get svc
NAME      TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
kubernetes  ClusterIP   10.96.0.1    <none>        443/TCP    119s
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prg7$ kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment  2/2     2            2          46s
```

